MARYLEBONE ROSTER · LINKS DESIGNER · THE SHORTLIST



Three good links, one edge each

Balanced V6, Level Crossing and Middle Ground, against each other and against today's link

Prepared 8 November 2026 · every figure is computed from the rotas

What all three share

Each line holds for all three, better than or level with today; page 2 has what is worse, and why these three.

✔ All **9 of 9** December staffing rules (today 4), and all three flexible ones: 14 on a Saturday, evenly spaced cover weeks, weekday closers from 15:45

✔ **Today's leave**: 14 days buy 30 days off at best, and four weeks off take 14 days' leave, as today

✔ **No two closing weeks side by side**, cover weeks aside, as today: a week of closing shifts is never followed by another

✔ **No avoidable fatigue warning** and never more than **five days in a row**, cover weeks aside (today 4 warnings and 7 days)

✔ The same **35-hour** week and **218.6 days** a year, under the ceiling of 219 (today 219.0)

✔ Every closing shift **shorter, by 15 minutes to an hour** — the weekday's and Sunday's 8h 10m on all three (today 8h 40m and 8h 55m); no shift over 9h 00m

Where they differ

the better of the three on each line in bold green; today for scale, not ranked

MEASURE	TODAY	BALANCED V6	LEVEL CROSSING	MIDDLE GROUND
Weeks on one shift time Mon–Fri, all earlies or all lates	7 of 16	7 of 21	**18 of 21**	**18 of 21**
Weeks mixing earlies and lates	7	3	**0**	**0**
Longest run of early or late weeks · with cover weeks	2 · 3	**3 · 5**	5 · 7	5 · 7
Week-to-week change of start	4h 00m	4h 08m	**1h 34m**	1h 42m
Single rest days, each a year	10	22	**10**	12
Full weekends off a year · longest wait, weeks	10 · 7	12 · 6	**14 · 5**	**14 · 5**
Worst cover week: fatigue warnings · days in a row	5 · 9	4 · 10	**0 · 6**	1 · 7
Heaviest week, Mon–Sat	43h 50m	42h 55m	**42h 30m**	**42h 30m**
Lightest week, Mon–Sat	25h 10m	24h 30m	**25h 30m**	**25h 30m**
Shortest rest between shifts, fixed duties	12h 30m	14h 20m	14h 20m	**14h 35m**
Closing shift, Saturday	9h 10m	8h 40m–8h 55m	8h 40m	**8h 10m**
Shifts over 8h 30m Mon–Sat a year · of them earlies	31 · 0	8 · 2	6 · **0**	**0 · 0**
Earlies finishing before 14:00, each a year	26	**20**	**20**	12
Shift times, and how many are new	18 (0 new)	17 (8 new)	**13 (6 new)**	**13 (6 new)**
New shift times on a Saturday	0	3	**2**	3
Fit to the trains, weekday · Sat · Sun (0 = exact)	52.0 · 65.9 · 70.2	40.0 · 17.3 · 29.2	**31.4** · 17.6 · 29.2	32.7 · **9.8** · **28.6**

Each one's edge

Balanced V6 MB-26-V6 · 1818a1c2

The shortest runs of earlies or of lates. At most three weeks of either in a row, five with cover weeks beside them, where the others have five and seven. Its cost: 7 of 21 weeks on one shift time, not 18, and three mixing earlies and lates; a 4h 08m change of start; 22 single rest days a year; 12 weekends off a year, not 14; up to four fatigue warnings and ten days in a row if a cover week falls badly.

Level Crossing LC-26-T10 · 32241e32

No fatigue warning in any cover-week placement tested, the steadiest start times. Never more than six days in a row however a cover week falls, where the others reach seven and ten; the gentlest change of start, 1h 34m; two 13:45s every weekday and five at the weekday open. Its cost: the most starts at 06:20, 58 a year each, the weakest Saturday fit (17.6), and 6 shifts a year over 8h 30m Monday to Saturday, where Middle Ground has none.

Middle Ground MG-26-T8 · 20668ac9

No shift over 8h 30m, the closest weekend fit. Alone of the three, nothing over 8h 30m on any day, and every closer 8h 10m; the longest rest, 14h 35m, the fewest starts at 06:20, 48 a year each, and no two 06:20 weeks side by side either. Its cost: the fewest early finishes before 14:00, 12 a year, not 20; four at the open every day; one fatigue warning if a cover week falls badly, where Level Crossing has none.

Which one, by what matters most

IF THE MAIN PRIORITY IS	CHOOSE
Of these three, the shortest runs of earlies or of lates	**Balanced V6**
Of these three, no fatigue warning in any cover-week placement tested	**Level Crossing**
Of these three, no shift over 8h 30m on any day, and the closest weekend fit	**Middle Ground**
OUTSIDE THE SHORTLIST	
A 16:00 closer Monday to Saturday, shorter than the three's	**Closing Time**
The fewest new shift times: three	**Home Grown**
A fourth person to the Saturday close	**Even Keel**
The steadiest start times from week to week	**Second Edition**

THE SHORTLIST · CONTINUED

Against today, and how to choose

What the three cost against today, why these three, and how to choose; a figure they share spans their columns

Against today's link a year: the rotation's count scaled to 52 weeks; days a year, full weekends off and the leave on page 1

MEASURE	TODAY	BALANCED V6	LEVEL CROSSING	MIDDLE GROUND
Saturdays a year, each	26		**28 on all three**	
Sunday overtime duties a year, each	21		**20 on all three**	
Starting at 06:20, each a year	62	50	58	48
Average shift, Mon–Sat	8h 14m		**8h 16m on all three**	

Why these three and not the other twenty-one

Shortlisted on 8 November 2026 for a new design target: no two closing weeks side by side, changes between earlies and lates placed at useful rest breaks, and weeks on one shift time still important; all three keep today's leave too. Crossover, first in the pack's league and 0.4 points ahead of Level Crossing, puts four closing weeks together. Balanced V6 scores far below every design but Balanced V5; against Level Crossing it loses most on the fatigue warnings a badly placed cover week can make, the weeks on one shift time and the weekends off. Each other design has something, and pays for it: **links-26-other-designs.pdf**, beside this sheet, sets the twenty-one side by side.

The honest costs, against today

- **Late finishes.** 42 a year each at 23:00 or later on all three (today 39), and 94 at 22:00 or later (today 78). Three to the close every day, where today has two Monday to Saturday, means no link meeting the rules has fewer than 42 at 23:00 or later.
- **Longer runs of earlies or of lates.** Up to five weeks in a row on Level Crossing and Middle Ground, seven with cover weeks beside them; three and five on Balanced V6 (today 2, up to 3).
- **Fewer long breaks.** Today five of its 17 rest-day breaks are three or four days, about 13 a year each; Balanced V6 has three, two of them four days, about 6 a year; Middle Ground two and Level Crossing one, each of three days.
- **Week to week, on Balanced V6.** 7 of its 21 weeks on one shift time, where today has 7 of 16; a 4h 08m change of start, against today's 4h 00m; and 22 single rest days a year, today 10 (Middle Ground 12, Level Crossing 10).
- **Early finishes.** Fewer before 14:00 than today's 26 a year each: 20 on Balanced V6 and Level Crossing, 12 on Middle Ground.
- **Saturdays.** About 2 more a year on all three (28, today 26): fourteen on a Saturday across 26 lines, today ten across 20.
- **New times.** 8 of Balanced V6's 17 shift times are new, 6 of Level Crossing's 13, 6 of Middle Ground's 13: shorter versions of today's lates on each (four, three and three), and besides, a 07:30–16:00 weekday early and the 9-hour Sunday early on Balanced V6 and Level Crossing; a 12:00–20:30 late and a 06:20–15:00 Saturday early on Balanced V6; a 06:20–14:45 Saturday early on Level Crossing; a 07:00–15:30 early and two Sunday earlies on Middle Ground.

How to choose each of the three is the one to choose for something different

- **Choose Balanced V6** if the shortest runs of earlies or of lates matter most — for 7 of 21 weeks on one shift time, a 4h 08m change of start, 22 single rest days a year and up to four fatigue warnings if a cover week falls badly.
- **Choose Level Crossing** if no fatigue warning in any cover-week placement tested, and steady start times, matter most — for the most starts at 06:20, 58 a year each, and the weakest Saturday fit of the three.
- **Choose Middle Ground** if no shift over 8h 30m and the closest weekend fit matter most — for the fewest early finishes, four at the open every day and one fatigue warning if a cover week falls badly.